

Lyu Muyang

PHD STUDENT · CENTER FOR LIFE SCIENCES

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Education

Peking University

Beijing, China

PHD INTEGRATIVE LIFE SCIENCES(PSYCHOLOGY)

2023 - present

- Advisor: Prof. Si Wu

Beijing Normal University

Beijing, China

BS COMPUTER SCIENCE AND TECHNOLOGY

2019 - 2023

- GPA: 3.8/4.0

Research Experience

Computational modeling of perceptual learning mechanisms based on two-stage neural networks

Peking University

ADVISOR: PROF. SI WU

Sep. 2022 - Present

Participating in the perceptual learning project and building two-stage neural networks for different characteristics of neural representation feature learning and task decision learning, respectively. The model reproduces more successfully the specificity and transfer properties of the classical psychophysical experiment of perceptual learning, the double-training experiment paradigm.

Large scale whole-brain simulation of zebrafish

Peking University

ADVISOR: PROF. SI WU

Sep. 2023 - Present

Participating in the project which aims to build a large scale whole-brain model of zebrafish combining with large-scale neural activity and whole-brain connectome data, and try to bridge the gap between the microstructural and dynamic attributes of the brain and the cognitive functions observed behaviorally.

Building model fitting module for BrainPy

Peking University

ADVISOR: PROF. SI WU

Dec. 2023 - Present

BrainPy is a highly flexible and extensible framework targeting on the general-purpose Brain Dynamics Programming (BDP) by Wu's lab. As a member of the development team, mainly involved in building model fitting module, fitting neurodynamic model hyperparameters to experimental data.

Flexible decision making program

Beijing Normal University

ADVISOR: PROF. ZONGLEI ZHEN

Jul. 2021 - Sep. 2022

Joined the lab of Prof. Zhen Zonglei, Department of Psychology, Beijing Normal University, Computational Neuroimaging Laboratory as a research assistant. Participated in the Flexible Decision Making Program in Qiyuan Lab. During the period, the research was about designing multi-brain region brain-like neural networks to solve the Raven's Progressive Matrices.

Hybrid smart home brain-computer interface system based on conceptual imagination and motor imagination

Beijing Normal University

ADVISOR: PROF.. XIA WU

Apr. 2021 - May 2022

Participated in the undergraduate research training and innovation and entrepreneurship Beijing-level project (Beijing Innovation) "Hybrid smart home brain-computer interface system based on conceptual imagination and motor imagination", supervised by Prof. Wu Xia of the School of Artificial Intelligence, Beijing Normal University. The project has been completed.

Research on path planning methods based on spatial cognitive mechanisms

Beijing Normal University

ADVISOR: PROF. BAILU SI

Apr. 2020 - May 2021

Led the undergraduate research training and innovation and entrepreneurship university-level project "Research on Path Planning Methods Based on Spatial Cognitive Mechanisms", supervised by Prof. Bailu Si from the School of System Science, Beijing Normal University. The project is about the simulation study on the characteristics of grid cell discharge. The project has been successfully completed.

Publications

PUBLISHED

Xiao Liu, **Muyang Lyu**, Cong Yu, Si Wu. Representation- and Task-based Plasticity Decides Perceptual Learning and its Specificity and Transfer: A Computational Model[J]. *Advances in Psychological Science*, 2023, 31(suppl.): 129-129.

Lyu Muyang, Ruixuan Liu, and Junyi Wang. "Solving raven's progressive matrices using rnn reasoning network." *2022 7th International Conference on Computational Intelligence and Applications (ICCIA)*. IEEE, 2022: 32-37.

Liu Ruixuan, **Muyang Lyu**, and Jiangrong Yang. "A hybrid BCI system combining motor imagery and conceptual imagery in a smart home environment." *2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 2022: 1066-1071.

Awards

- 2021 **Meritorious Winner of Mathematical Contest In Modeling**, the Consortium for Mathematics and its Applications
- 2020 **First Prize of Contemporary Undergraduate Mathematical Contest in Modeling in Beijing**, China Society for Industrial and Applied Mathematics
- 2020 **Jingshi First Class Scholarship (top 10%)**, Beijing Normal University
- 2021 **Jingshi Second Class Scholarship (top 20%)**, Beijing Normal University
- 2021 **Scholarship for the second class of the competition**, Beijing Normal University
- 2022 **Jingshi Third Class Scholarship (top 30%)**, Beijing Normal University

Skills

Programming: Python (Pytorch, BrainPy, Brian2), R, MATLAB

Languages: English (CET6 589); Mandarin (native); Japanese (N2)